

Circuit Board Description by Zones

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The circuit board is divided into **eight functional zones**, each serving a specific purpose in controlling and monitoring the ceiling lights.

Breadboard Notes

- **Breadboard connections:** All five pins on either side of the central divider of the breadboard are electrically connected to one another.
- **Arduino pin numbering:** The breadboard schematic uses the **ATmega/ATmel pin numbers**. The corresponding **Arduino board pin numbers** are shown in parentheses.

Zone 1 — IR Receiver

Zone 1 contains the **infrared (IR) receiver**.

The output signal from the handheld IR controller is received by the IR receiver and sent to **Arduino pin 2**.

This signal is used to command the circuit to turn the ceiling lights on or off.

A 47uF Radial Capacitor, 16V, is connected across the IR Receiver Ground and Power terminals to filter and stabilize the power supply.

Zone 2 — Transistor Control

Zone 2 contains a **transistor that receives a control signal from the ATmega/Arduino and controls the electromechanical (EM) switch**.

Arduino pin 3 sends a control signal through **R3**, which commands the ceiling lights on or off through the EM switch.

Zone 3 — Power Divider

Zone 3 is the **power distribution section** of the circuit.

A single **12-volt power supply** provides power for both the circuit board and the ceiling-lights.

The power divider distributes the appropriate supply voltage to the various sections of the circuit.

Zone 4 — Electromechanical Switch and Voltage Divider

Zone 4 contains the **electromechanical (EM) switch and a voltage divider**.

The voltage divider serves several purposes:

1. It provides a **control point for the transistor in Zone 2**.
2. It provides a **connection point for the EM switch**.
3. It provides a **voltage test point** that is monitored by the voltage comparator in Zone 5.

The voltage at this test point represents the state of the ceiling-light circuit and is used by the comparator to determine when the lights have been turned off.

Zone 5 — Voltage Comparator and Transistor

Zone 5 contains the **voltage comparator and a transistor**.

The voltage comparator monitors the voltage from the **test point in Zone 4** and compares it against a **reference voltage**.

When the **test voltage** falls **below the reference voltage**, the comparator changes state and causes the ceiling lights to turn off.

When the **reference voltage is pulled to ground by Arduino pin 4**, the reference voltage falls below the test voltage. This causes the circuit to return to its original state and reset.

Zone 6 — Reference Voltage Adjustment

Zone 6 is used to **set and adjust the reference voltage** used by the comparator in Zone 5.

This reference voltage establishes the threshold at which the comparator determines that the ceiling lights have been turned off.

Zone 7 — Circuit Status Indicator

Zone 7 provides a **visual indication of the circuit's operating state** using an LED.

Arduino pin 5 monitors the output voltage level of the voltage comparator. This allows the Arduino to monitor the comparator's state and determine the operating condition of the circuit.

LED ON — Circuit Enabled

When the LED is **on**, the circuit is enabled and ready to receive commands from the IR controller.

If the ceiling lights are on and the IR controller commands them to turn off, the circuit will initiate the process of turning the lights off.

If the lights are **already off** and the IR controller sends an OFF command, the circuit will still attempt to turn the lights off. Since they are already off, nothing will change.

After the **15-second debounce timer** completes, the circuit will reset. The LED will turn back on, indicating that the circuit is once again enabled and ready for the next command.

LED OFF — Lights Being Controlled OFF

When the LED is **off**, it indicates that the circuit is controlling the ceiling lights in the **OFF state**.

The LED will illuminate again when the circuit receives the appropriate command from the IR controller or when the wall switch is turned off, causing the circuit to return to its enabled/reset state.

Zone 8 — Electromechanical Switch Control

Zone 8 controls the **electromechanical (EM) switch**.

It also contains a **diode that protects the rest of the circuit from the voltage spike generated by the EM switch's internal coil** when the coil is switched off.

The **output of the voltage comparator in Zone 5** controls the EM switch through this section of the circuit.

Overall Circuit Function

Together, the eight zones form a control and monitoring system for the ceiling lights.

The **IR receiver** accepts commands from the handheld controller. The **Arduino** processes those commands and controls the transistor and electromechanical switch. The **voltage divider and comparator** monitor the resulting electrical state of the lighting circuit. Once the desired state is detected, the comparator and Arduino work together to reset the circuit, while the **LED provides a visual indication of whether the circuit is enabled or controlling the lights off**.

The **15-second debounce period** prevents the circuit from immediately responding to transient or unstable signals and provides a controlled reset before the system is ready for the next command.